

Java Inheritance & Polymorphism

1. Inheritance (Single & Multilevel)

1. Create a class `Person` with name and age. Inherit it in class `Employee` and add salary. Write a method to display details from both classes.
2. Create a class `Animal` with a method `makeSound()`. Inherit it in class `Dog` and override `makeSound()` to print "Bark".
3. Create a class `A`, inherit it in `B`, then `B` in `C` (Multilevel). Add one method in each class and call all from `C` object.
4. Write a program with base class `Vehicle` and derived class `Car` that adds a method `fuelType()`. Show single inheritance.
5. Demonstrate multilevel inheritance with classes `Shape` → `Rectangle` → `ColoredRectangle`. Call methods of all classes using `ColoredRectangle` object.

2. Polymorphism (Compile-time & Run-time)

1. Create a class `MathOperation` with two `add()` methods (overloading): one for int, another for double. Show compile-time polymorphism.
2. Create a class `Printer` with method `print(String)`, `print(int)`, and `print(double)`. Demonstrate method overloading.
3. Create a class `Animal` with method `sound()`. Inherit in class `Cat` and override `sound()` to print "Meow". Show run-time polymorphism using a parent reference.
4. Write a class `Bank` with method `getInterestRate()`. Inherit it in `SBI`, `ICICI`, override the method in each. Call using parent reference to show runtime polymorphism.
5. Create class `Shape` with method `area()`. Override it in `Circle` and `Square`. Call using `Shape` reference to different objects.